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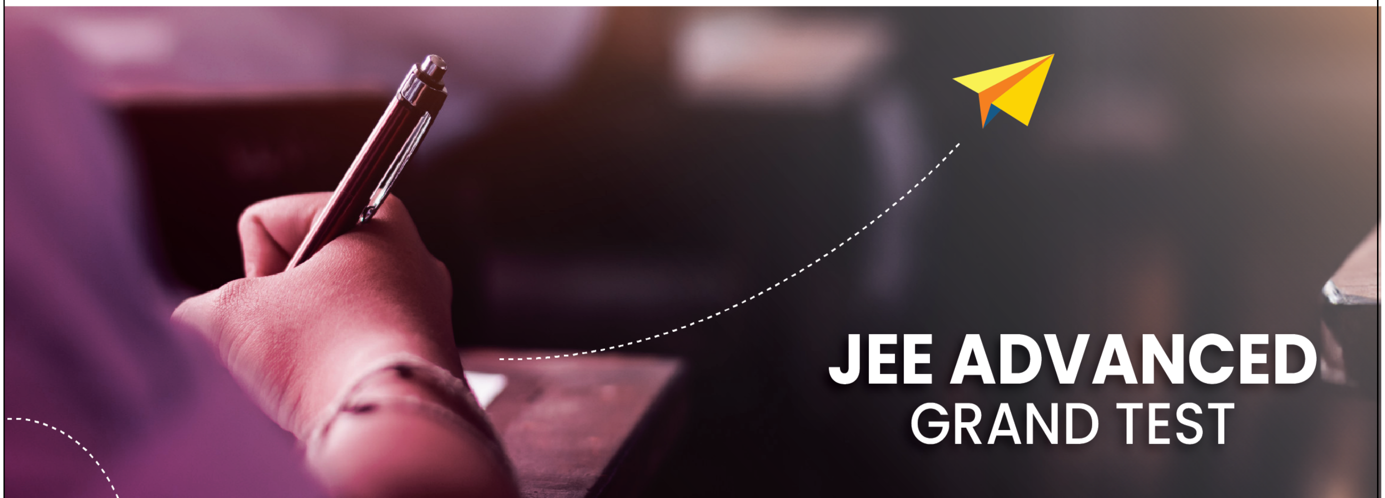

Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_Elite, Target & LIIT-BTs	**Paper -1(Adv-2017-P1-Model)**	**Date: 15-05-2024**
Time: 09.00Am to 12.00Pm	**GTA-33**	**Max. Marks: 183**

15-05-2024_Sr.Super60_Elite, Target & LIIT-BTs_Jee-Adv(2017-P1)_GTA-33_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: ____________________ **H.T. NO:** | | | | | | | |

JEE-ADVANCE-2017-P1-Model

Time: 03:00 Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 183

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 8 – 12)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 13 – 18)	3 COLUMNS MATCHING With Single Answer Type	+3	-1	6	18
Total				18	61

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 26 – 30)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 31 – 36)	3 COLUMNS MATCHING With Single Answer Type	+3	-1	6	18
Total				18	61

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 44 – 48)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 49 – 54)	3 COLUMNS MATCHING With Single Answer Type	+3	-1	6	18
Total				18	61

PHYSICS

Max. Marks:61

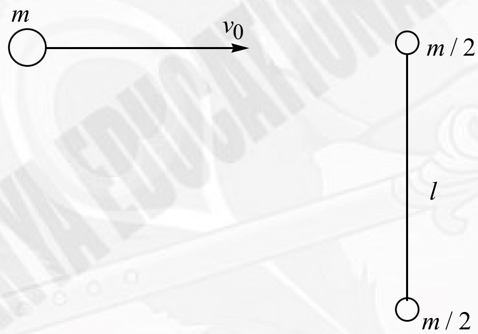
SECTION -1 (Maximum Marks: 28)

- This Section Contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) Correct.
- For each question, darken the bubble(s) corresponding to all the correct options(s) in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

Full Marks : +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks -2 In all other cases.

For example, if (A), (C)and (D) are all the correct options for a question, darkening all these three will get +4 marks; darkening only (A) and (D) will get +2 marks; and darkening (A) and (B) will get -2 marks; as a wrong option is also darkened

1. A ball of mass m moving with velocity v_0 experiences a head-on elastic collision with one of the spheres of a stationary rigid dumbell resting on a smooth surface as shown in figure. The mass of each sphere equals $\frac{m}{2}$, and the distance between them is l.
Disregarding the size of the spheres, which of the following statements are correct



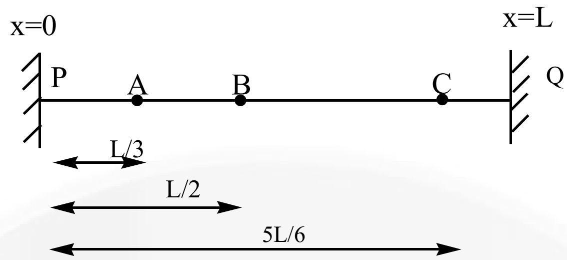
A) Angular momentum of dumbbell after collision in the reference frame moving translationally and fixed to the dumbbell's centre of inertia of the system $= \frac{1}{3}mv_0 l$

B) Angular speed of dumbell after collision $\omega = \frac{4v_0}{3l}$

C) A frame fixed to the centre of mass of the dumbbell is not an inertial reference frame.

D) Speed of centre of mass v_{cm} of the dumbbell and angular speed ω of dumbell after collision are related as $v_{cm} = \frac{l}{2}\omega$

2. Standing waves are established on a string of length L such that A is a node and B is an immediate antinode. Oscillation amplitude for point B is a_0. Let a' be the oscillation amplitude for point C. Pick the suitable option (s) for correct value of a' and the possible equation (s) for standing waves on the string.



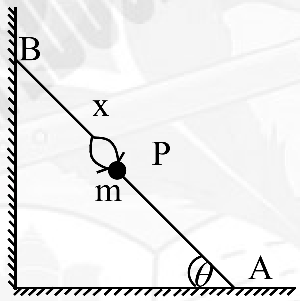
A) $a' = \dfrac{a_0}{2}$

B) $y = a_0 \sin\left(\dfrac{6\pi}{L}x\right)\cos(\omega t)$

C) $a' = a_0$

D) $y = a_0 \sin\left(\dfrac{3\pi x}{L}\right)\sin(\omega t)$

3. A light stick of length l rests with its one end against the smooth wall and other end against the smooth horizontal floor as shown in the figure. The bug starts at rest from point B and moves such that the stick always remains at rest. a_P is the magnitude of acceleration of bug of mass m, which depends upon its distance of x from the top end of the stick. Choose the correct option(s)



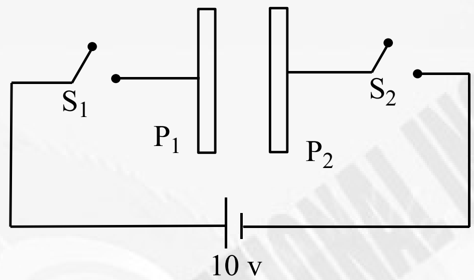
A) $a_P = \dfrac{g}{\sin\theta}\left(1 - \dfrac{x}{\ell}\right)$

B) $a_P = \dfrac{g}{\cos\theta}\left(1 - \dfrac{x}{\ell}\right)$

C) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\dfrac{\pi}{2}\sqrt{\dfrac{\ell \sin\theta}{3g}}$

D) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\dfrac{\pi}{2}\sqrt{\dfrac{\ell \sin\theta}{g}}$

4. Two identical plates P_1 and P_2, of cross sectional area A are separated at a very small distance d, where $\frac{\varepsilon_0 A}{d} = 2$ F. Initially charge on the plate P_1 is -10 C and charge on the plate P_2 is +30 C. Now plate P_1 and P_2 are connected to positive and negative terminals of battery of potential difference 10 Volts with the help of switches S_1 and S_2 respectively. Choose the correct option (s) when S_1 & S_2 are closed



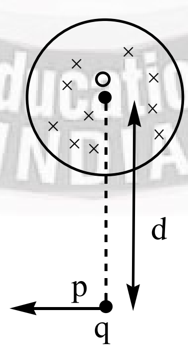
A) The work done by the battery is 200 J.

B) The net charge on the plate P_2 is -10 C

C) The charge on the left side of the plate P_1 is 10C.

D) The charge on the right side of the plate P_2 is 20C.

5. In the cylindrical region of radius R = 2m, there exists a time varying magnetic field B such that $\frac{dB}{dt} = 2$ Tesla/sec. A charged particle having charge q = 2C is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to line OP by an external agent up to infinity so that there is no gain in kinetic energy of charged particle. Then



A) Work done by external agent is 4π Joule when d = 4m

B) Work done by external agent is 4π Joule when d = 8m

C) Work done by external agent is independent of d.

D) Work done by external agent is positive.

6. Mark out the correct statement (s)

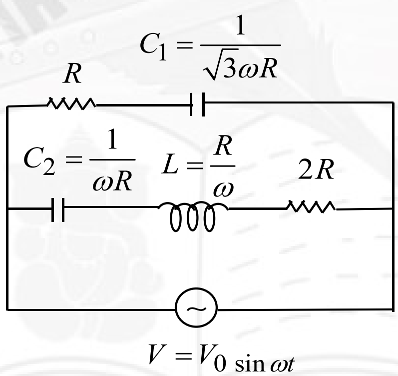
A) In alpha decay, the energy released is shared between alpha particle and daughter nucleus in the form of kinetic energy and share of alpha particle is more than that of the daughter nucleus.

B) In beta β^- decay, the energy released is shared between electron and antineutrino.

C) A nuclide undergoes α decay and reach ground state, then all the α particles emitted by that nuclide will have almost the same speed.

D) A nuclide undergoes β decay then all the β particles emitted by that nuclide may have widely different speeds.

7. Consider the AC circuit shown and choose correct alternative (s).



A) Effective reactance of the circuit is $\sqrt{3}R$.

B) Effective impedance of the circuit is $\frac{2R}{\sqrt{3}}$.

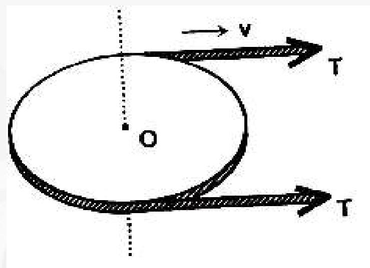
C) Current supplied by AC source as a function of time is. $\frac{\sqrt{3}V_0}{2R}\sin\left(\omega t + \frac{\pi}{6}\right)$

D) Effective wattles current flowing in the circuit is $\frac{\sqrt{3}V_0}{4R}$.

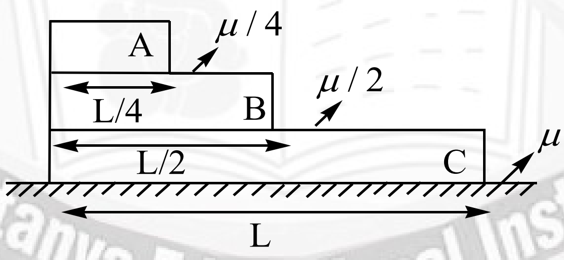
SECTION -2 (Maximum Marks: 15)

- This section Contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the **ORS.**
- For each question, marks will be awarded in one of the following categories:
  Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
  Negative Marks: 0 In all other cases

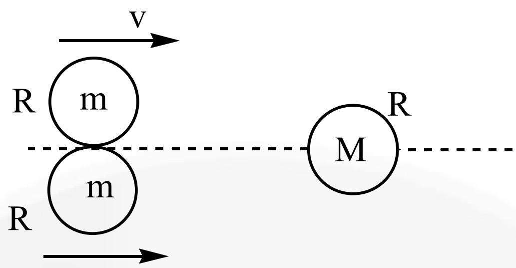
8. A flexible drive belt runs over a frictionless pulley as shown in figure. The pulley is rotating freely about the vertical axis passing through the centre O of the pulley. The vertical axis is fixed on the horizontal smooth surface. The mass per unit length of the drive belt is 1 kg/m and the tension in the drive belt 8N. The speed of the drive belt is 2 m/s. Find the net normal force applied by the belt on the pulley in newton.



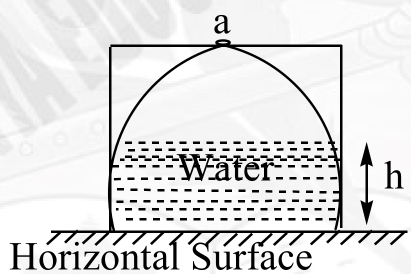
9. Three blocks A,B and C of mass m, $\frac{m}{2}$ and m of different densities and dimensions are placed over each other as shown in the figure. The coefficients of friction are shown. Blocks placed in a vertical line are made to move towards right with same velocity at the same instant. Find the time (in sec) taken by the upper block A to topple from the middle block B. Assume that blocks B and C don't stop sliding before A topples from B. (given L = 36 m, $\mu = 0.4$ and $g = 10 m/s^2$)



10. Two identical discs of mass m and of radius R touch each other and move with the same velocity perpendicularly to the line segment which joins their centres of mass, along the surface of a horizontal smooth tabletop. There is a third disc of mass M and of radius R at rest, at a point on the perpendicular bisector of the line segment joining the centres of mass of the two moving discs as shown in the figure. The two moving discs collide elastically with the third one, which is at rest. There is no friction between the rims of the discs. What should the ratio of M/m be in order that after the collision the two discs of mass m move perpendicularly to their initial velocity?



11. Consider a cuboidal vessel $(2R \times 2R \times R)$ with a hemispherical cavity of radius R, kept at a horizontal smooth surface as shown in the figure. The vessel has very small hole of cross – sectional area "a". Now water of density ρ is poured into space developed due horizontal surface and vessel through hole very slowly. When the height of water level is h, the vessel lifts off the surface and liquid leaks through space generated. (R = 2m, $\rho = 10^3\ kg/m^3$, m=mass of the vessel $= 9\pi\ kg$). If the value of h is 5k cm, find the value of k. (Atmospheric pressure = $10^5\ N/m^2$).



12. A 2 kg block moving with 10 m/sec strikes a spring of constant π^2N/m which is attached to 2 kg block at rest kept on a smooth horizontal floor. Find the time (in sec) for which rear moving block remain in contact with spring.

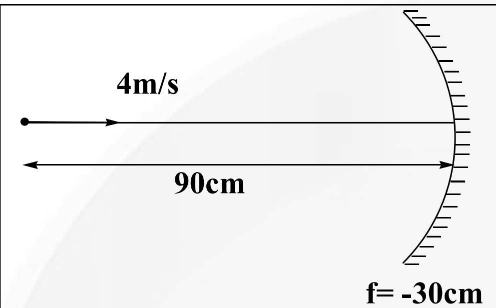
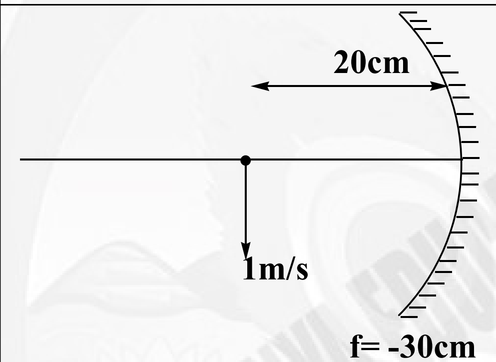
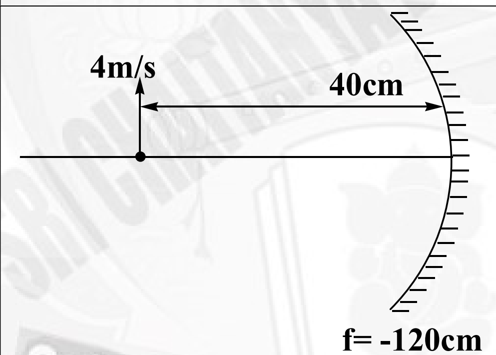
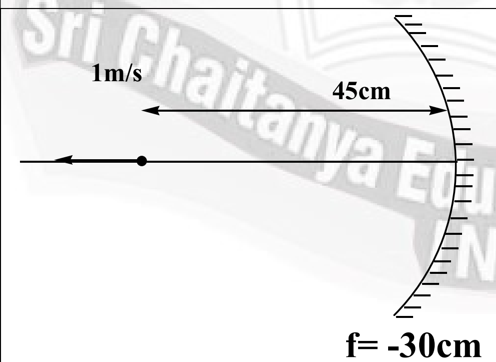
SECTION -3 (Maximum Marks: 18)

- This section Contains **SIX** questions of matching type.
- This section Contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the **ORS**.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened
Negative Marks : -1 In all other cases

Answer question 13,14 and 15 by appropriately matching the information given in the three columns of the following table.

Column – I lists four situations showing a concave mirror and a moving point object. Column – II lists the corresponding image coordinate (usual sign convention) and column – III lists the image velocity.

Column I		**Column 2**		**Column 3**	
(I)		(i)	v = +60 cm	(P)	$+6\hat{j}$
(II)		(ii)	v = -45 cm	(Q)	$-\hat{i}$
(III)		(iii	v = -120 cm	(R)	$-3\hat{j}$
(IV)		(iv	v = -90 cm	(S)	$+4i$

13. Pick the correct option.

A) $(IV)(iii)(P)$ B) $(III)(iv)(Q)$ C) $(II)(i)(R)$ D) $(I)(i)(S)$

14. Pick the correct option for maximum magnitude of image coordinate

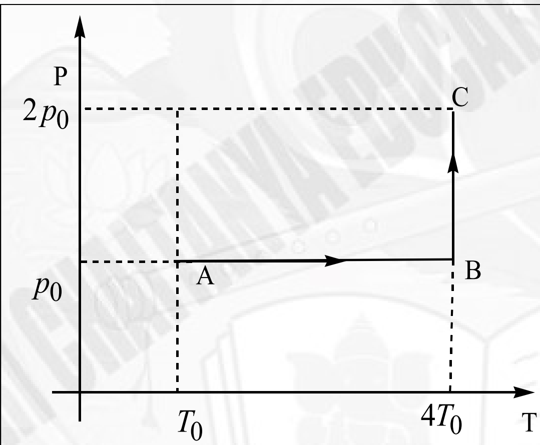
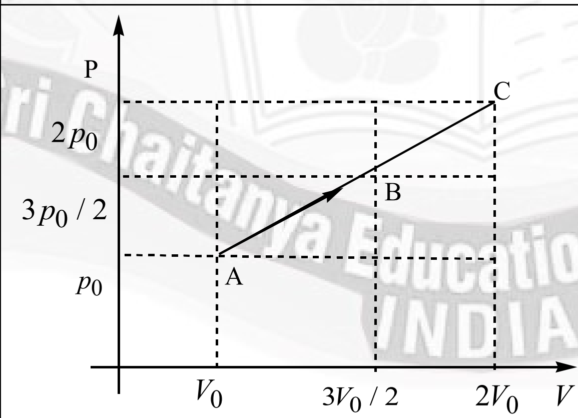
A) $(I)(i)(S)$ **B)** $(II)(i)(R)$ **C)** $(III)(iv)(Q)$ **D)** $(IV)(iv)(S)$

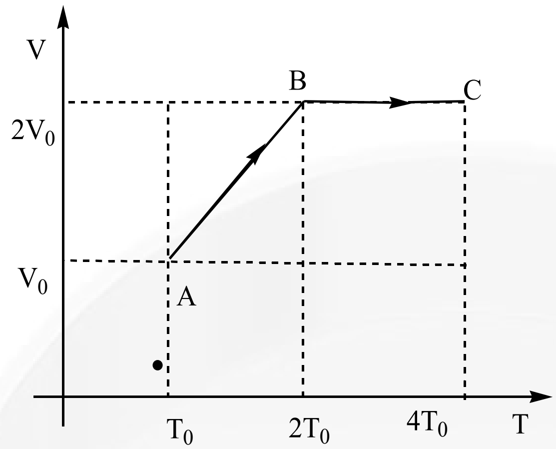
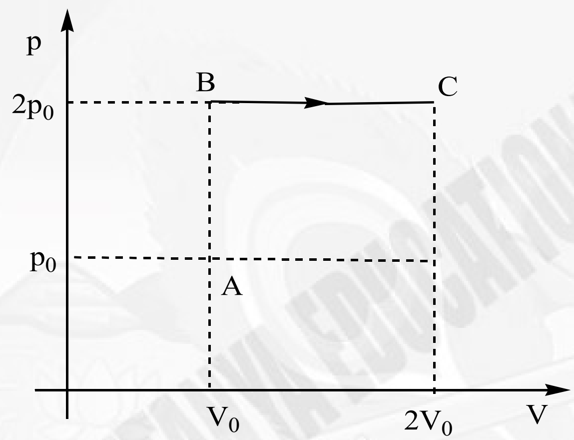
15. Pick the correct option for maximum magnitude of image velocity.

A) $(III)(i)(P)$ **B)** $(II)(iii)(R)$ **C)** $(I)(iv)(S)$ **D)** $(IV)(iii)(Q)$

Answer question 16, 17 and 18 by appropriately matching the information given in the three columns of the following table.

Column – I shows 2 moles of a monoatomic gas being taken from initial state $A(p_0, v_0, T_0)$ to final state $C(2p_0, 2v_0, 4T_0)$ via an intermediate state B. Column II and column III list the values of work done by the gas in process $A \rightarrow B$ and $B \rightarrow C$ respectively.

	Column I		Column 2		Column 3
(I)		(i)	$2RT_0$	(P)	$\frac{7}{4}RT_0$
(II)		(ii)	$6RT_0$	(Q)	zero

(III)		(iii)	zero	(R)	$-8RT_0\ell n(2)$
(IV)		(iv)	$\frac{5}{4}RT_0$	(S)	$4RT_0$

16. Pick the correct option.

A) $(I)(i)(S)$ **B)** $(II)(iv)(P)$ **C)** $(III)(iii)(R)$ **D)** $(IV)(ii)(Q)$

17. Pick the correct option for minimum value of work done in process $A \to B$.

A) $(IV)(iii)(S)$ **B)** $(III)(iii)(R)$ **C)** $(II)(iv)(P)$ **D)** $(I)(i)(S)$

18. Pick the correct option for minimum value of work done in process $B \to C$.

A) $(I)(ii)(R)$ **B)** $(II)(iv)(P)$ **C)** $(III)(iii)(R)$ **D)** $(IV)(ii)(Q)$

CHEMISTRY Max.Marks:61

SECTION -1 (Maximum Marks: 28)

- This Section Contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) Correct.
- For each question, darken the bubble(s) corresponding to all the correct options(s) in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

Full Marks : +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks : +1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks -2 In all other cases.

For example, if (A), (C)and (D) are all the correct options for a question, darkening all these three will get +4 marks; darkening only (A) and (D) will get +2 marks; and darkening (A) and (B) will get -2 marks; as a wrong option is also darkened

19. The correct set of statements among the following

A) Micelles formation takes place above CMC and above Kraft temperature

B) Larger the length of carbon chain of surfactant lower the CMC at a given temperature

C) Addition of excess of electrolyte to sol suppresses the diffused double layer and reduces zeta potential there by causes coagulation

D) Ultra-microscope is based on Tyndall effect

20. Which of the following statements is/are CORRECT?

A) 1 mole Benzene $\left(P^{\circ}_{Benzene}=42mm\right)$ and 2 mole Toluene $\left(P^{\circ}_{Toluene}=36mm\right)$ liquid mixture will have mole fraction of benzene in vapour phase equal to 7/19.

B) If two liquids that from an ideal solution are mixed, the change in entropy is positive.

C) Higher the value of K_H (Henry's law constant) at a given pressure and temperature, lower is the solubility of the gas in the liquid.

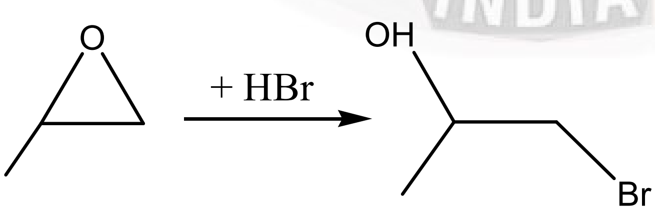
D) Solubility of a gas in liquid increase with increase in temperature at a given pressure.

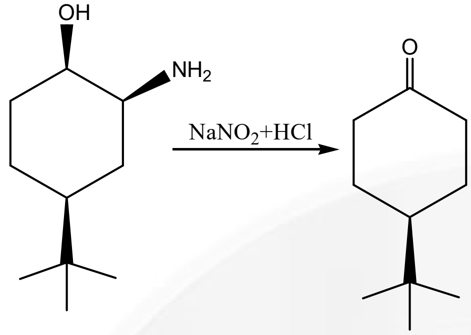
21. In which of the following reaction(s) correct major product given ?

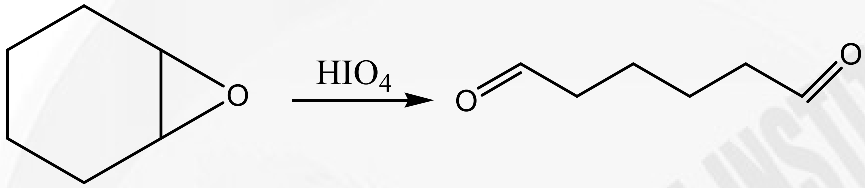
A)

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-O-CH_3\xrightarrow{conc.HCl}CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH+CH_3Cl$$

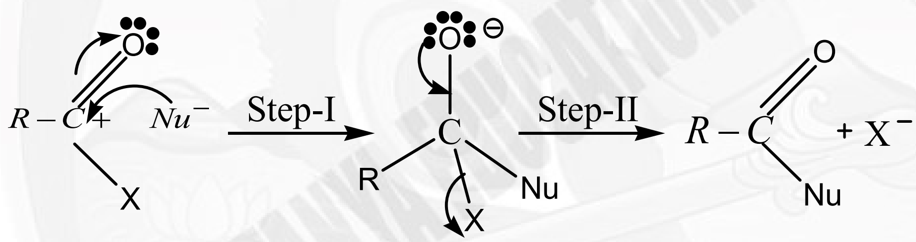
B)



C) 

D) 

22. General mechanism for nucleophilic acyl substitution reactions of carboxylic acid derivatives is given below



For base catalysed hydrolysis of amides which is nucleophilic acyl substitution reaction, correct statement is/are

A) Possible on prolonged heating with strong base.

B) Step- II is slower than step-I

C) At moderate concentration of base order of reaction is 2

D) When concentration of base is very high , order of reaction is 3

23. In which of the following reaction(s) $POCl_3$ is obtained as one of the product?

A) $PCl_5 + P_4O_{10} \xrightarrow{(6:1)}$ **B)** $PCl_5 + H_2O \xrightarrow{(1:1)}$

C) $PCl_5 + SO_2 \xrightarrow{(1:1)}$ **D)** $PCl_3 + C_2H_5OH \xrightarrow{(1:3)}$

24. Consider the following reaction and identify the correct statement(s) for the balanced equation?

$$Au + \text{Aqua regia} \longrightarrow X_{(aq)} + Y\uparrow + H_2O$$

[Given that: 'Y' is a diatomic molecule]

A) 'X' is diamagnetic and 'Y' is paramagnetic

B) 'Y' is paramagnetic with two unpaired electrons

C) 1 mole Au reduces 3 moles of NO_3^-

D) Aqua regia is a mixture of 1 part Conc.HNO_3 and 3 parts Conc. HCl by volume.

25. Which of the following statement(s) is/ are correct?

A) Among Zn, Cr, Cu and V; Zn has highest first ionization enthalpy

B) Among Zn, Mn, Fe and V; Zn has lowest enthalpy of atomization

C) Both the first and second ionization enthalpies are more for Ti than V

D) Among Ni, Co and Fe; density is highest for Ni

SECTION -2 (Maximum Marks: 15)

- This section Contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

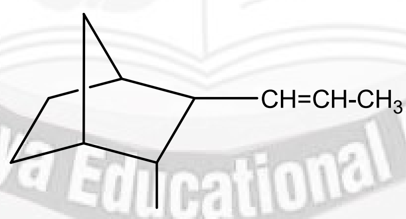
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Negative Marks: 0 In all other cases

26. The reaction $2A+B\rightarrow C+D$ follows the rate law $-\frac{d[B]}{dt}=K[A][B]^2$. Calculate the value of $(x+y)$?

S.no	$(A)_0\times10^{-4}M$	$(B)_0\times10^{-4}M$	Half life (sec)
1.	500	2	10
2	500	4	x
3.	2	250	8
4.	4	500	y

27. Total number of stereo isomers possible for the following compound is x. $\frac{x}{4}$ value is



28. The formula of alkyl silicon chloride for different types of silicones are given below. For straight chain silicones: $Si(R)_{x_1}(Cl)_{y_1}$

For cross-linked silicones: $Si(R)_{x_2}(Cl)_{y_2}$

Then (y_1+y_2) is_________.

29. How many of the following liberate coloured vapours/gas/fumes with conc. H_2SO_4 on or before heating?

i) $KCl(s)+K_2Cr_2O_7(s)$, **ii)** $KNO_2(s)$, **iii)** $KI(s)$, **iv)** $K_2S(s)$, **v)** $KCl(s)$,

vi) $KBr(s)+MnO_2(s)$, **vii)** $KNO_3(s)$, **viii)** $KCl(s)+MnO_2(s)$, **ix)** $K_2SO_3(s)$

30. Total number of diamagnetic complexes from the following:

(i) $\left[Co(H_2O)_6\right]^{2+}$ **(ii)** $\left[Co(H_2O)_6\right]^{3+}$ **(iii)** $\left[Co(ox)_3\right]^{3-}$ **(iv)** $\left[Co(H_2O)_3F_3\right]$

(v) $\left[Co(NH_3)_6\right]^{2+}$ **(vi)** $\left[Co(NO_2)_6\right]^{4-}$ **(vii)** $\left[Co(NO_2)_6\right]^{3-}$

SECTION -3 (Maximum Marks: 18)

- This section Contains **SIX** questions of matching type.
- This section Contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened
Negative Marks : -1 In all other cases

Answer Q.31, Q.32 and Q.33 by appropriately matching the information given in the three columns of the following table.

S.no	Column-I	Column-II	Column-III
I	Irreversible Adiabatic expansion of an ideal gas	i) $\Delta S_{sys}=+ve$	P) $\Delta S_{total}=0$
II	Free expansion of an ideal gas	ii) $\Delta S_{sys}=0$	Q) $\Delta S_{total}=+ve$
III	Melting of a substance at its Boiling point	iii) $\Delta G=0$	R) $\Delta S_{total}=-ve$
IV	Vaporisation at Boiling point	iv) $\Delta G=-ve$	S) Isoentropic process

31. The only correct combination is

A) $I-ii-S$ **B)** $I-i-S$ **C)** $II-ii-Q$ **D)** $II-i-Q$

32. The only correct combination is-

A) $III-iv-Q$ **B)** $III-iii-P$ **C)** $IV-i-Q$ **D)** $IV-iii-R$

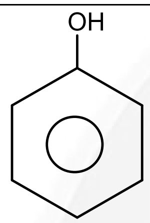
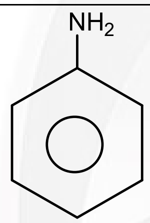
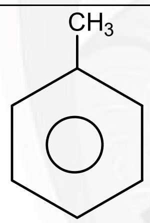
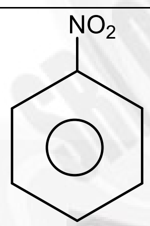
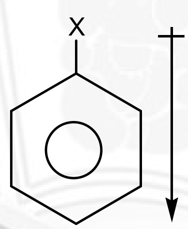
33. The only correct combination is-

A) $I-i-Q$ **B)** $II-iii-P$ **C)** $III-i-P$ **D)** $IV-i-R$

Answer Q.34, Q.35 and Q.36 by appropriately matching the information given in the three columns of the following table.

Answer the following three questions by using List-I,II and III

(A.E.S means aromatic electrophilic substitution)

List-I		List-II		List-III	
P)		I)	In A.E.S at meta position less reactive than C_6H_6	a)	Diazo coupling (AES) possible with $C_6H_5N_2Cl$ at pH=4-5 (or) 9-10
Q)		II)	In A.E.S at ortho, meta and para position more reactive than benzene	b)	Fridel craft alkylation ($R-Cl/AlCl_3$) possible
R)		III)	In A.E.S at ortho, meta and para position less reactive than C_6H_6	c)	Nitrosoation (AES) possible with $NaNO_2 + HCl$
S)		IV)	Net dipole moment direction is 	d)	Bromination (AES) possible without any Lewis acid catalyst

34. Among the following correct match is

A) P-II-C **B)** Q-I-C **C)** R-IV-d **D)** Q-I-d

35. Among the flowing incorrect match is

A) R-II-b **B)** R-IV-a **C)** S-III-b **D)** S-III-c

36. Among the following correct match is

A) P-I-C **B)** P-II-d **C)** R-II-C **D)** Q-II-a

MATHEMATICS Max.Marks:61

SECTION -1 (Maximum Marks: 28)

- This Section Contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) Correct.
- For each question, darken the bubble(s) corresponding to all the correct options(s) in the **ORS.**
- For each question, marks will be awarded in one of the following categories:
  Full Marks : +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
  Partial Marks : +1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
  Zero Marks : 0 If none of the bubbles is darkened.
  Negative Marks -2 In all other cases.

For example, if (A), (C)and (D) are all the correct options for a question, darkening all these three will get +4 marks; darkening only (A) and (D) will get +2 marks; and darkening (A) and (B) will get -2 marks; as a wrong option is also darkened

37. Let $f(x)=\cos^{-1}\left(\cos\left(\dfrac{24+4x^2}{4+x^2}\right)\right)$ and $f:A\to B$, where A is the set of all possible values of x for which $f(x)+5-2\pi\le 0$, then which of the following is/are TRUE

A) number of integral points (x, y) lying on the line $9x+\alpha^2 y=16$; α is an integer and $x\in A$, are 8

B) area bounded by y=f(x) and x-axis is $6\pi-16$

C) range of f(x) is $[2\pi-6, 2\pi-5]$

D) range of f(x) is $[2\pi-6, 2\pi-4)$

38. Let P(0, 3, 4) and Q(5, 12, 0) be two given points. A is a point on x-axis for which (PA+QA) is minimum. B is a point on y-axis for which (PB+QB) is minimum. C is a point on z-axis for which for which (PC+QC) is minimum, then which of the following statements is/are TRUE?

A) Equation of plane passing through A, B, C is $(1547)x+(325)y+(700)z=2275$

B) Equation of plane passing through A, B, C is $(1364)x+(325)y+(1050)z=2275$

C) The volume of tetrahedron OABC (where O is origin) is $\dfrac{2275}{144}$ unit^3

D) The volume of tetrahedron OABC (where O is origin) is $\dfrac{2275}{408}$ unit^3

39. Let P(x) be a monic quadratic polynomial with rational coefficients and

$P\left(\sqrt{5+\sqrt{5+\sqrt{5+........\infty terms}}}\right)=0$ then:

A) The value of $2024\sum_{r=2}^{2024}\frac{1}{P(r)+5}=2023$ **B)** $\lim_{n\to\infty}\sum_{r=3}^{n}\frac{1}{p(r)+3}=1$

C) $\int_{-1}^{1}\frac{dx}{P(x)+x+6}=\frac{\pi}{2}$ **D)** $\int_{1}^{2}\frac{dx}{\sqrt{P(x)+x+4}}=\ln\left(2+\sqrt{3}\right)$

40. Suppose function f(x) is continuous and differentiable at x=0, and function g(x) is continuous but non-differentiable at x=0. If the function $h(x)=e^{g(x)}f^{2}(x)g(x)$ is differentiable at x=0, then which of the following is/are CORRECT?

A) f(0) must be equal to 0 **B)** g(0) must be any positive value

C) f(0) = 0 or g(0) = -1 **D)** f(0) = g(0)

41. Which of the following is/are correct?

A) If A is a $n\times n$ matrix such that $a_{ij}=\left(i^{2}+j^{2}-5ij\right)(j-i)\forall$ i,j then trace $(A)=0$

B) If A is a $n\times n$ matrix such that $a_{ij}=\left(i^{2}+j^{2}-5ij\right)(j-i)\forall$ i,j then trace $(A)\neq 0$

C) If P is a 3×3 orthogonal matrix, and α,β,γ are angles made by a straight line with X, Y, Z axes respectively and if $A=\begin{bmatrix}\sin^{2}\alpha & \sin\alpha\sin\beta & \sin\alpha\sin\gamma\\ \sin\alpha\sin\beta & \sin^{2}\beta & \sin\beta\sin\gamma\\ \sin\alpha\sin\gamma & \sin\beta\sin\gamma & \sin^{2}\gamma\end{bmatrix}$, $Q=P^{T}$ AP,

then $PQ^{6}P^{T}$ =32A

D) If matrix $A=\left[a_{ij}\right]_{3\times 3}$ and matrix $B=\left[b_{ij}\right]_{3\times 3}$ where $a_{ij}+a_{ji}=0$ and $b_{ij}-b_{ji}=0\forall$ i and j then $A^{28}B^{11}$ is a singular matrix

42. Let $g(x)=ax^2+bx+c$, where $a,b,c\in N$ and $\int_0^1 g(x)dx=\frac{11}{6}$. Let $f(x)$ be a continuous and derivable function in (x_1,x_2). If $f(x).f'(x)\geq x\sqrt{1-(f(x))^4}$ and $\lim\limits_{x\to x_1^+}(f(x))^2=1$ and $\lim\limits_{x\to x_2^-}(f(x))^2=\frac{1}{2}$, then minimum value of $\left[x_1^2-x_2^2\right]$ is…… (where [.] denotes greatest integer function)

A) 2 - abc **B)** a + b - c **C)** b +c **D)** a + b

43. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that the angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that

$|\vec{c}|=4,\ \vec{c}\times\vec{b}=2\vec{a}\times\vec{b}\ \ and\ \ \vec{c}=\lambda\vec{a}+\mu\vec{b}$ then,

A) $\lambda=2$

B) Sum of values of $\mu=-1$

C) Product of all possible values of $\mu=-12$

D) Number of distinct values of μ are three.

SECTION -2 (Maximum Marks: 15)

- This section Contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Negative Marks: 0 In all other cases

44. If the number of solution(s) of the equation

$(10+6\sqrt{3})\sin^3 x+(6\sqrt{3}-10)\cos^3 x+6\sin 2x=8$ which satisfying

$|\tan x|>1, x\in[0,2027\pi]$ are N, then the sum of the digits of N is

45. The radius of the circle which passes through the focus of the parabola $x^2=4y$ and touches it at the point (6, 9) is $K\sqrt{10}$, then K=

46. Consider the locus of the complex number z in the Argand plane given by $\operatorname{Re}(z)-2=|z-7+2i|$.Let $P(z_1),Q(z_2)$ be two complex numbers satisfying the given locus and also satisfying $\arg\left(\dfrac{z_1-(2+ai)}{z_2-(2+ai)}\right)=\dfrac{\pi}{2},(a\in R)$ .If the minimum value of PQ is K then the value of $\left[\dfrac{K}{3}\right]$ is….. (where [.] is the step function)

47. Let a be a real number. If the value of definite integral $\int\limits_{-\pi+a}^{3\pi+a}|x-a-\pi|\sin\left(\dfrac{x}{2}\right)dx=-16$ such that the sum of all the values of a in [0 , 314] is $k\pi$, then the sum of the digits of k is

48. Let p, q, r are prime numbers and α,β,γ are positive integers such that LCM of α,β,γ is p^3q^2r and greatest common divisor of α,β,γ is pqr. If the number of possible triplets (α,β,γ) will be K, then number of divisors of K which are of the form 3n, $n\in N$ are

SECTION -3 (Maximum Marks: 18)

- This section Contains **SIX** questions of matching type.
- This section Contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options [A], [B], [C], and [D]. **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the **ORS.**
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened
Negative Marks : -1 In all other cases

Answer Q.49, Q.50 and Q.51 by appropriately matching the information given in the three columns of the following table.

Match for the curves given in column-I with the number of common tangents in column-II and with corresponding equation of common tangent in column-III.

	Column-I		Column-II		Column-III
(I)	Let $S_1 = x^2 + y^2 - 2x = 0$ and $S_2 = x^2 + y^2 - 12x + 32 = 0$	(i)	number of common tangent is 4	(P)	equation of common tangent is x=0
(II)	Let $S_1 = x^2 + y^2 - 2x = 0$ and $S_2 = 5x^2 + 5y^2 - x - 12y - 24 = 0$	(ii)	number of common tangent is 3	(Q)	equation of common tangent is $3x - 4y - 8 = 0$
(III)	Let $S_1 = x^2 + y^2 - 12x + 20 = 0$ and $S_2 = x^2 + y^2 - 6x = 0$	(iii)	number of common tangent is 2	(R)	equation of common tangent is $4y + 3x - 8 = 0$
(IV)	Let $S_1 = x^2 + y^2 - 6x = 0$ and $S_2 = x^2 + y^2 - 2x = 0$	(iv)	number of tangent is 1	(S)	equation of common tangent is $2\sqrt{2}y - x + 12 = 0$

49. Which of the following combination is **CORRECT?**

A) (I)(ii)(R) **B)** (I) (i) (S) **C)** (I)(i)(Q) **D)** (I) (ii) (Q)

50. Which of the following combination is **CORRECT?**

A) (II)(ii)(R) **B)** (II) (iii) (Q) **C)** (II)(iii)(R) **D)** (II) (iv) (Q)

51. Which of the following combination is **CORRECT?**

A) (I)(i)(R) **B)** (III)(iii)(S) **C)** (II)(iii)(Q) **D)** (IV)(ii)(P)

Answer Q.52, Q.53 and Q.54 by appropriately matching the information given in the three columns of the following table.

Match the condition of column-I with corresponding number of real roots of $f(x) = 0$ in column-II and number of points of non-derivability of $y = |f(|x|)|$ in column-III,

Where $f(x) = ax^2 + bx + c$, $a, b, c \in R$

Column-I		Column-II		Column-III	
(I)	$a^2+b^2+c^2-ab-bc-ca\le 0,\ a+b+c\ne 0$	(i)	0	(P)	0
(II)	$a^2+b^2+c^2+ab+bc+ca\le 0$	(ii)	1	(Q)	1
(III)	$3\left(a^2+b^2+c^2+1\right)\le 2\left(a+b+c+ab+bc+ca\right)$	(iii)	2	(R)	3
(IV)	$a^2+b^2+c^2\le 2a+6b+4c-14$	(iv)	∞	(S)	5

52. Which of the following is **correct** combination?

A) (I) (i) (P) **B)** (I) (iii) (Q) **C)** (II) (iv) (P) **D)** (III) (ii) (Q)

53. Which of the following is **incorrect** combination?

A) (III) (i) (Q) **B)** (II) (i) (P) **C)** (I) (i) (Q) **D)** (II) (iv) (P)

54. Which of the following is **correct** combination?

A) (III) (i) (P) **B)** (II) (i) (R) **C)** (IV) (iii) (R) **D)** (IV) (iii) (Q).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